CS4804-C26 : Final Project

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Project Book

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S&P 500 Market Analysis

How has the market changed in key areas of Stocks over the last 10 years

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Overview

Our project aimed to gain a deeper understanding of the behavior of the stock market by analyzing different visual trends of the S&P 500. In our dashboard, we discussed four different aspects of the market's behavior: price, returns, volatility, and drawdowns. To do this, we had to answer a few questions.

1. How has price changed over time?
2. How has market changed daily in percentage?
3. How risky has the market been over the years?
4. How far has it fallen from the peak price every day?

These different aspects combined allowed us to gain a deeper understanding of the market and the risks associated with it

Motivation

The motivation behind this project was a result of our recent interest in getting a better understanding of how the financial market operates. Our team is very curious about investing, and we realized that we don't have a core understanding of how the market is influenced. Our project provided us with the opportunity to explore those core concepts by analyzing data and different visual trends. In the end, we hope to develop a strong understanding of financial markets, which would allow us to make more informed decisions about investing in stocks

Related Work

Since we are seniors and will soon be entering the workforce, we have started thinking more about financial planning and investment opportunities. One of the most common investment benchmarks is the S&P 500, which represents the performance of many of the largest companies in the United States. Investopedia was founded in 1999 and provides a lot of educational content on the basics of investing.

S&P 500 Index: What It's for and Why It's Important in Investing

By WILL KENTON Updated November 20, 2025

Reviewed by MICHAEL J BOYLE

Fact checked by BETSY PETRICK

Evolution of our Questions

Initial Questions:
In the beginning of the project, we were focused on exploring and answering some general questions about the way the stock market behaves. We started with simple ideas like how the price of the market behaves over time, and how much it moves on any given day, and these drove our understanding general nature of the data and the market in general.

Expanding our Questions:
As we started poking around with the dataset and developing some basic graphs, we realized that these changes in price could be used to help us understand deeper questions about the market. From this, we started to focus a bit more on risk and market stability questions, spending time looking at volatility and drawdown, which provide us the ability understand periods in the market that it does not behave as simply as in just a basic price move

Deeper Questions:
At the end of the project, our questions shifted a bit again as we worked to try and get at why the market behaves in certain ways. Instead of just looking at time periods where moves in price, volatility, and drawdown, and understanding just 'how' the market behaves, but help us think about non-data-related causes for these shifts, and trying to get at the underlying basis for moves in the market.

Our Final Questions We Aim to Answer

Question 1: How has the price of the market changed over time?

Learning about the price changes is effective in teaching us how a stock market can fluctuate, and seeing trends by seeing long-term trends

Question 2: How does the market change on a daily percentage basis?

The daily percentage change gives us insight into short-term movements. This allows us to see how frequently the market experiences losses and or gains.

Question 3: How risky has the market been over the years

Volatility allows us to measure how unpredictable the market can be. It’s a key concept in the financial market because it reflects the level of risk associated with fluctuations.

Question 4: How far has it fallen from the peak price every day?

Learning about the differences between a peak and current price and how much it can change over time. Drawdowns highlight periods of market downturns and provide insight into the magnitude of losses investors may experience

What Our Data is

The data set that we used in this project included historical S&P index prices over time. Each row in the data represents the index value on a specific day.

The data set contains two main fields:

- **Date**
  - Represents the trading day
- **S&P 500 price**
  - Represents the closing value of the index on that day

During our preprocessing phase, we cleaned the data set by parsing dates into a time format, removing missing values, and sorting the data chronologically in Excel.

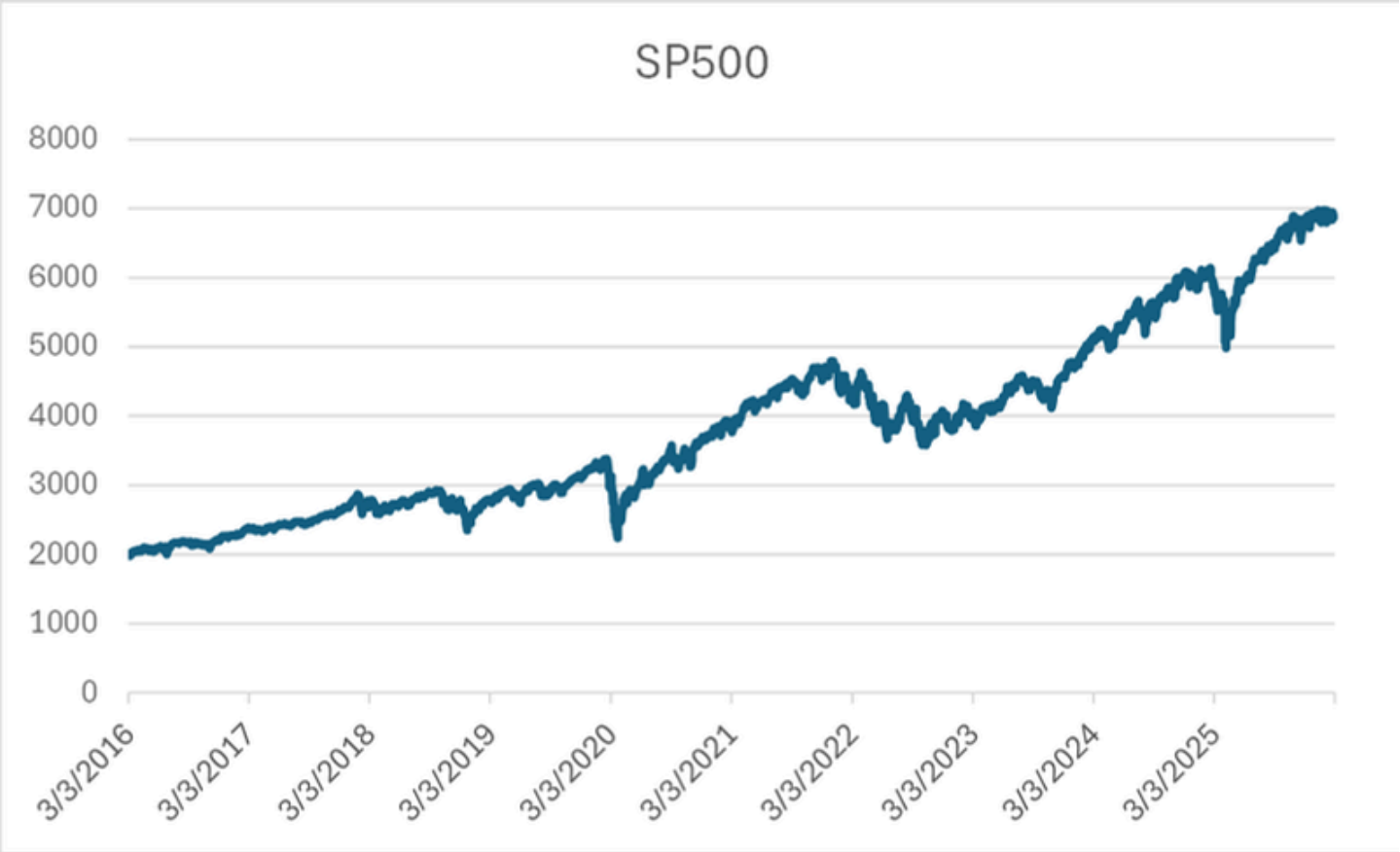
From this cleaned dataset, we were able to derive several metrics

1. **Daily returns**: The percentage change in price between days
2. **Rolling Volatility**: A 30-day standard deviation of returns
3. **Moving Averages**: A 50 and 200-day averages
4. **Drawdowns**: The percent decline from the previous peak price

Exploratory Data Analysis

Before building the final dashboard, we first wanted to plot the S&P price over time in Excel, which revealed a couple of patterns to us:

- The market **increased** over a long period of time
- The sharp declines correspond to different **economic crises**
- The market **recovers** eventually from those crises



Exploratory Data Analysis

From there, we decided to calculate how much the market moves over time using a specific **Return formula**.

Below is an example of how the **daily percentage** of change was calculated using an example value of 4000 for today and 4040 for yesterday.

$$\text{Return} = \frac{\text{Price}_t}{\text{Price}_{t-1}} - 1$$

Yesterday's Price = \$4000

Today's Price = \$4040

Return = (4040 / 4000) - 1 = 0.01

↳ 1% increase in value

This helped us gain insight into the distribution of market **gains** and **losses**.

Exploratory Data Analysis

We also explored market **volatility**, which shows how **unstable or risky** the market is. The volatility measures **how big price swings are.**

We calculated the standard deviation in Excel to see how spread out the returns are, which showed us two key insights:

- If the returns are positive → volatility is **low**
- If the returns are negative → volatility is **high**
- During **calm** market periods, volatility is **low**, but during **crises,** it increases **dramatically**

Exploratory Data Analysis

Finally, we examined **drawdowns,** which measure the **percentage drop** from the highest price previously reached

First, we track the **highest price ever reached** using a simple MAX function in Excel. From there, we used a Drawdown formula to compute the drawdown percentages. Below is an example of the calculation we did using fake peak and price numbers

$$\text{Drawdown} = \frac{\text{Price}}{\text{Peak}} - 1$$

Peak = $4500

Current = $4050

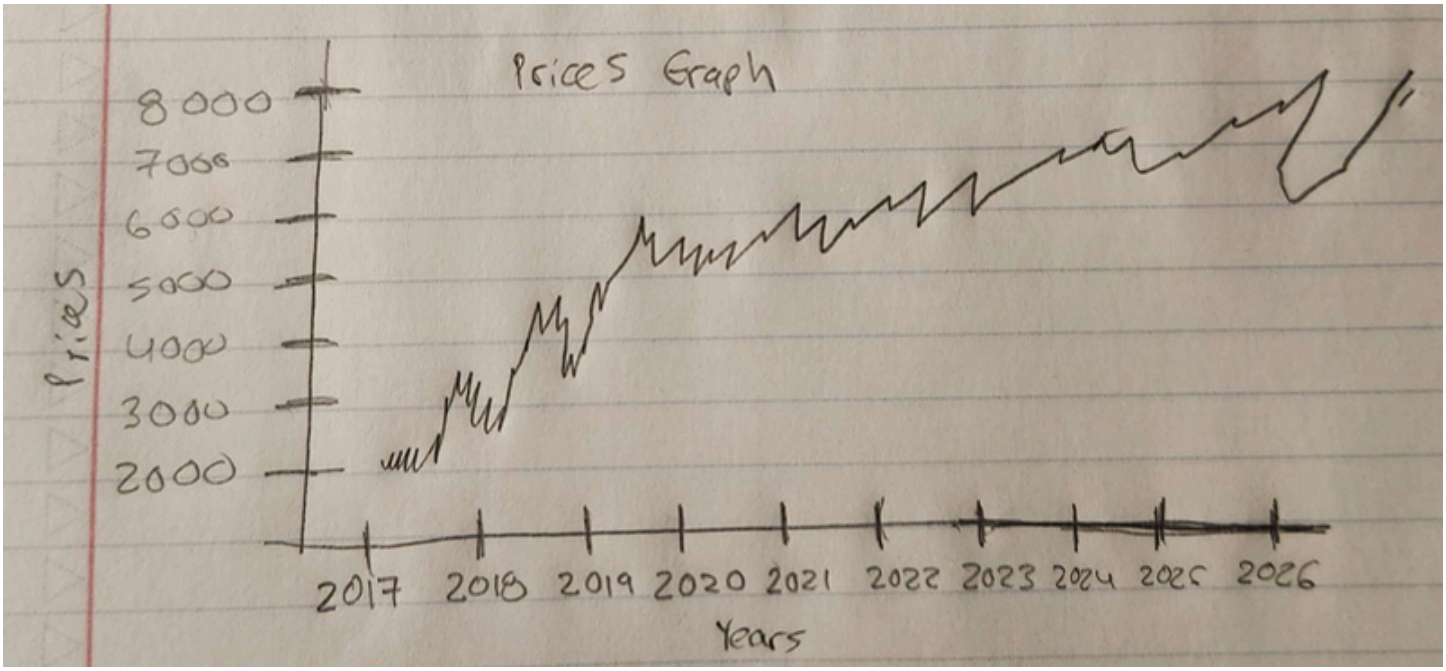
Drawdown = (4050/4500) - 1 = -0.1

↳ -10% drawdown

This helped identify major **downturns** and market **crashes**

Initial Design

Before we decided on the different aspects of our dashboard, we decided to start with a single price chart to capture the market. Below shows our initial sketch



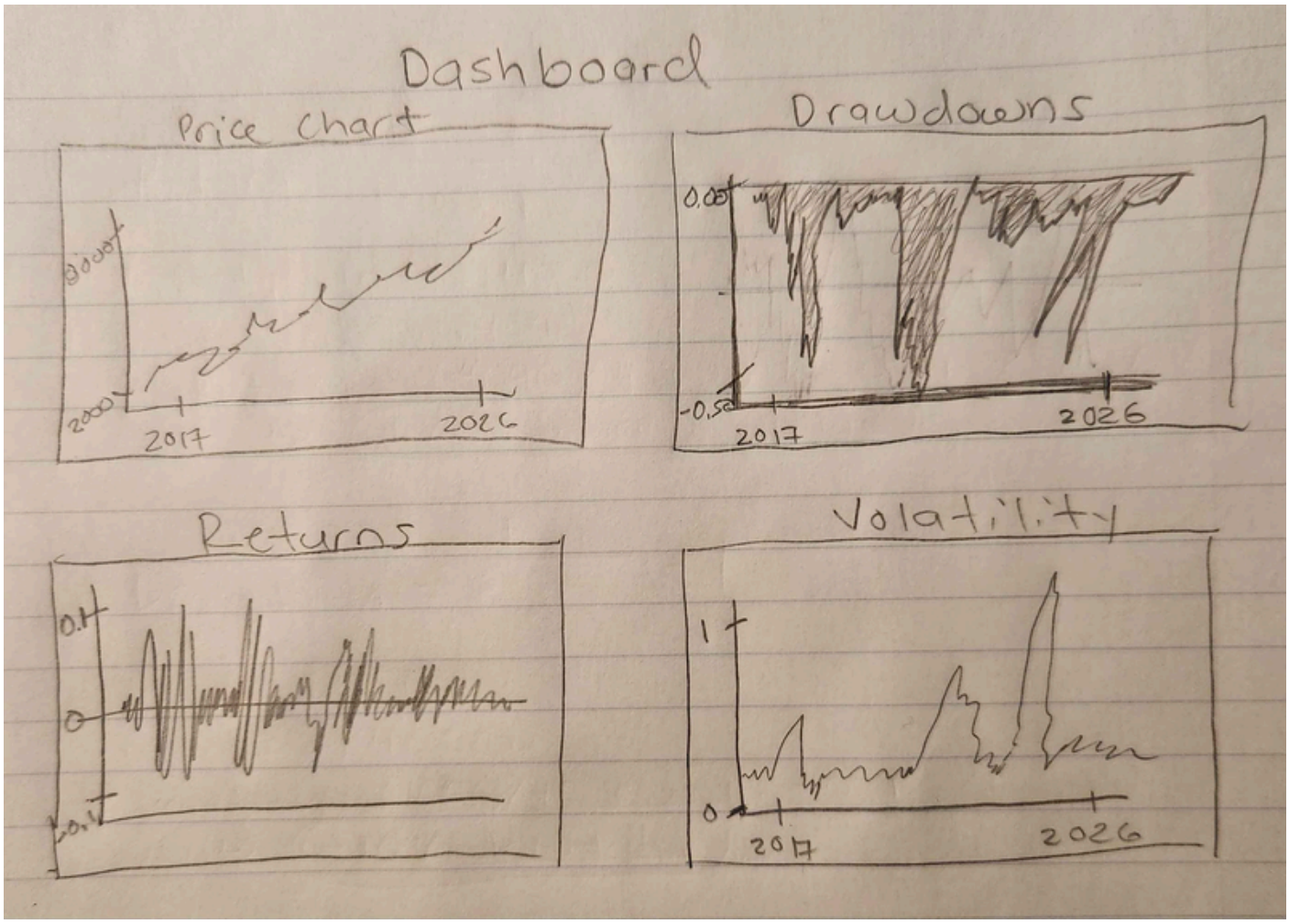
However, we quickly realized that it wasn't feasible because it didn't capture some of the more niche aspects.

- Instead of just price, we wanted to capture how the market moved on a day-to-day basis, so we decided to calculate the different return values.
- Additionally, we wanted a way to show how random the market can be in a smaller window, so we used volatility to measure the price swings
- Finally, we also wanted a way to show the different losses from the market's highest-value peaks, so we used drawdown to show how far the market has fallen from its previous peak

In the end, we would have a dashboard containing a Price Chart, Returns Chart, Volatility Chart, and Drawdown Chart. This design allows users to understand multiple aspects of the market simultaneously.

Expanding on Our Design

Below is the initial sketch of our dashboard



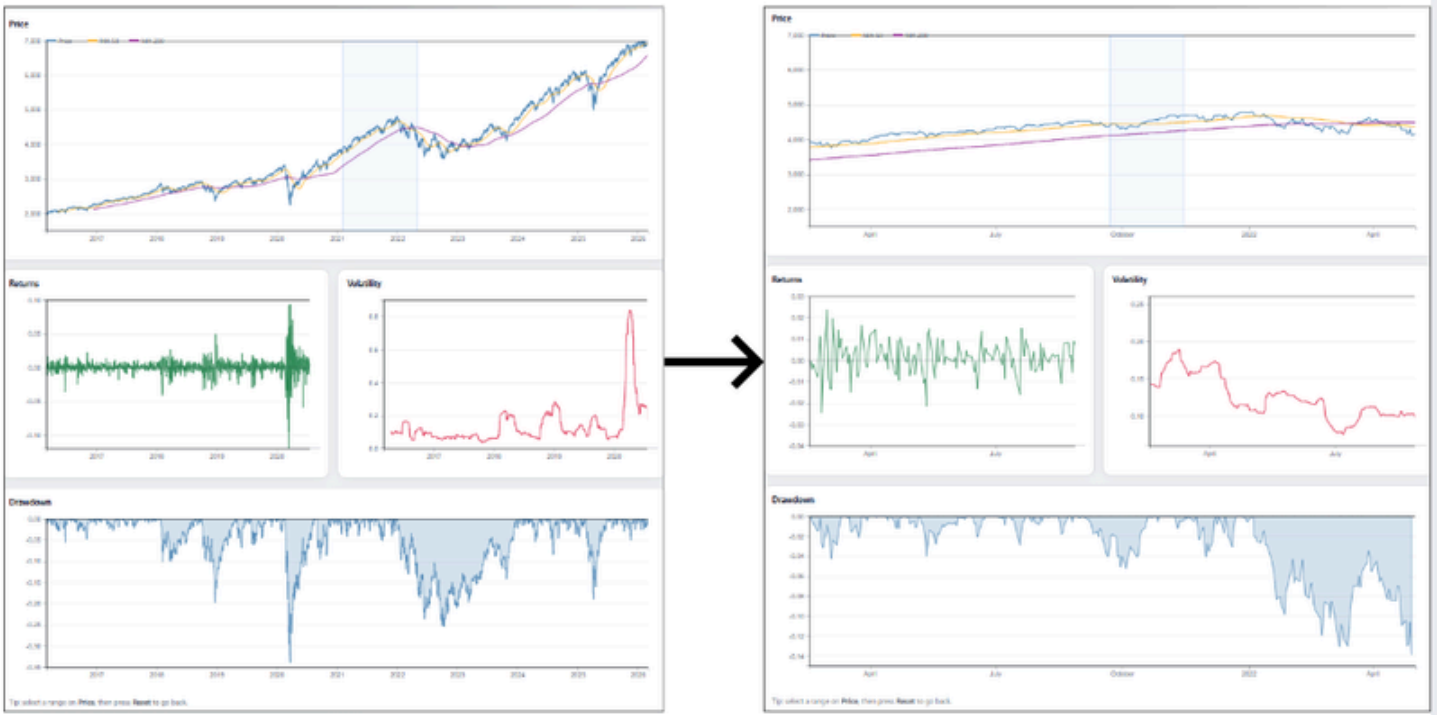
It showed how we wanted to capture the essence of our dashboard, but it didn't showcase any of the elements we attached alongside it. We wanted to make the dashboard interactive and show extra information that isn't as easily accessible. We also wanted to expand some of the graphs, like Price Chart and Drawdowns Chart, to really show the spread of the data more smoothly.

Final Design

This final dashboard design uses a clean, card-based layout to clearly organize the information while ensuring that the interface is visually simple and easy to navigate. We show the header at the top to provide the dashboard title, interaction instructions, and control elements like a reset button and display toggle. The main part of the dashboard arranges the visuals using a structured grid layout that creates a visual hierarchy that balances the structure. Each of the sections is contained within a rounded card with small, consistent spacing and shadows. These ensure a visual appeal to the human eye while maintaining a cohesive design. The goal of this design is readability, an intuitive interface, and visual appeal

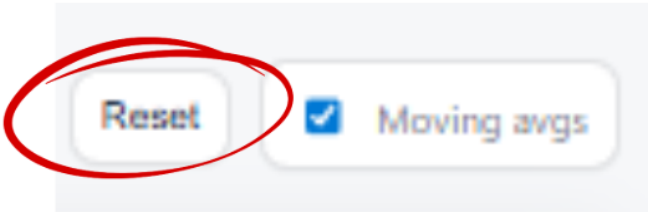
Zooming Feature

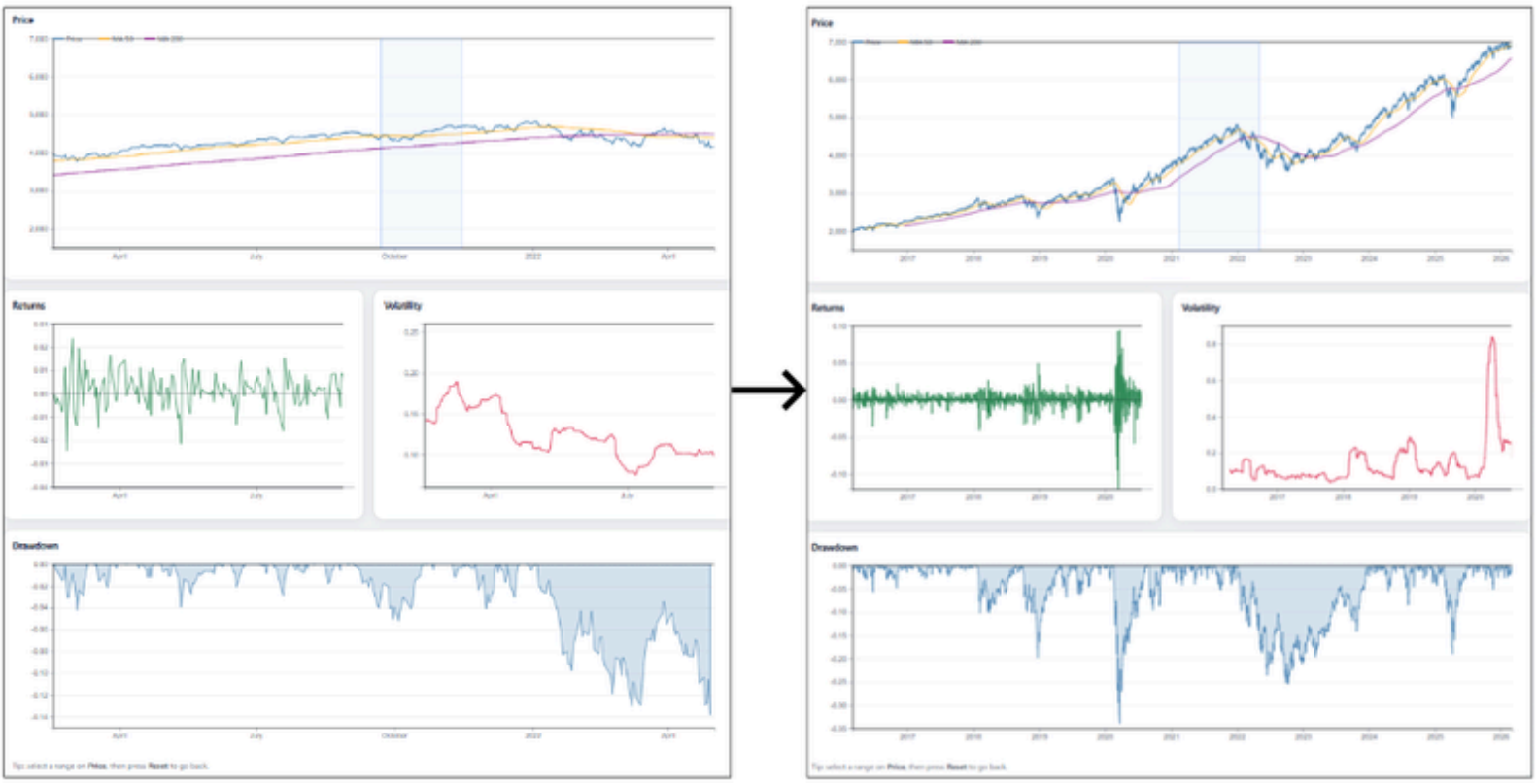
We implemented an interactive zooming feature, where people can select a time range in the price chart and automatically update the other charts to match that range. This way, users can focus on specific time periods and analyze how different metrics behaved during those periods.



Reset Feature

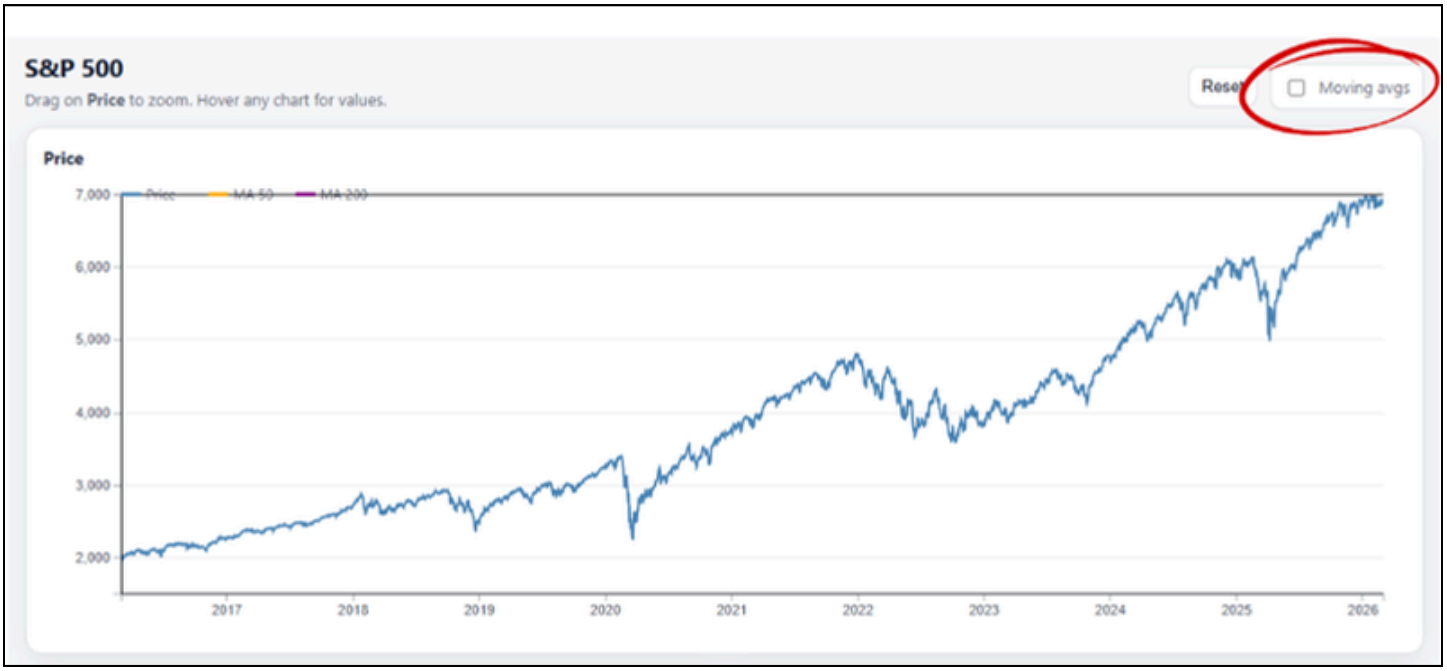
We also have a reset button at the top that resets the user's selected range to the original range. This ensures a smooth transition back to the original state of the dashboard





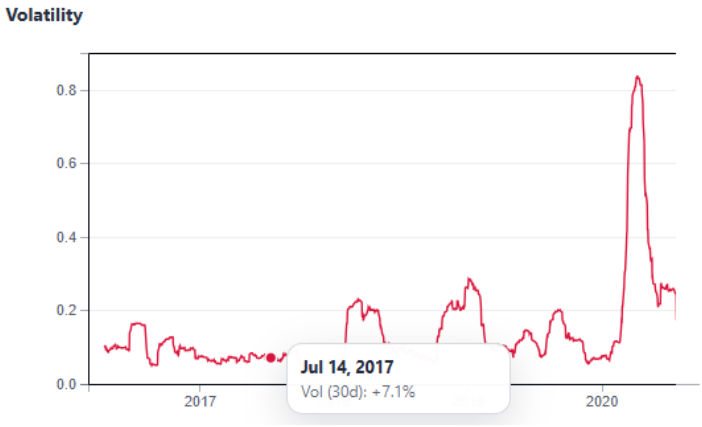
Graph Adjustment Feature

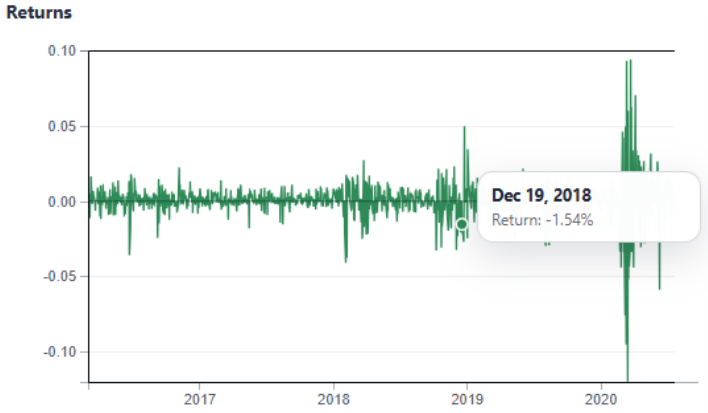
Next to the reset button is a checkbox called “moving args”. Once unchecked, it will show the original price chart graph without the other gold and purple lines. This allows us to visualize the raw price chart change over time without seeing the 50-day and 200-day averages



Hover Features

Finally, the dashboard includes the ability to hover over the different charts to display a tooltip with information about the selected point. It shows the corresponding date that is highlighted as well as the calculated percentage. The interactions allow the user to inspect the values on the graph rather than relying on just the visual trend of the line

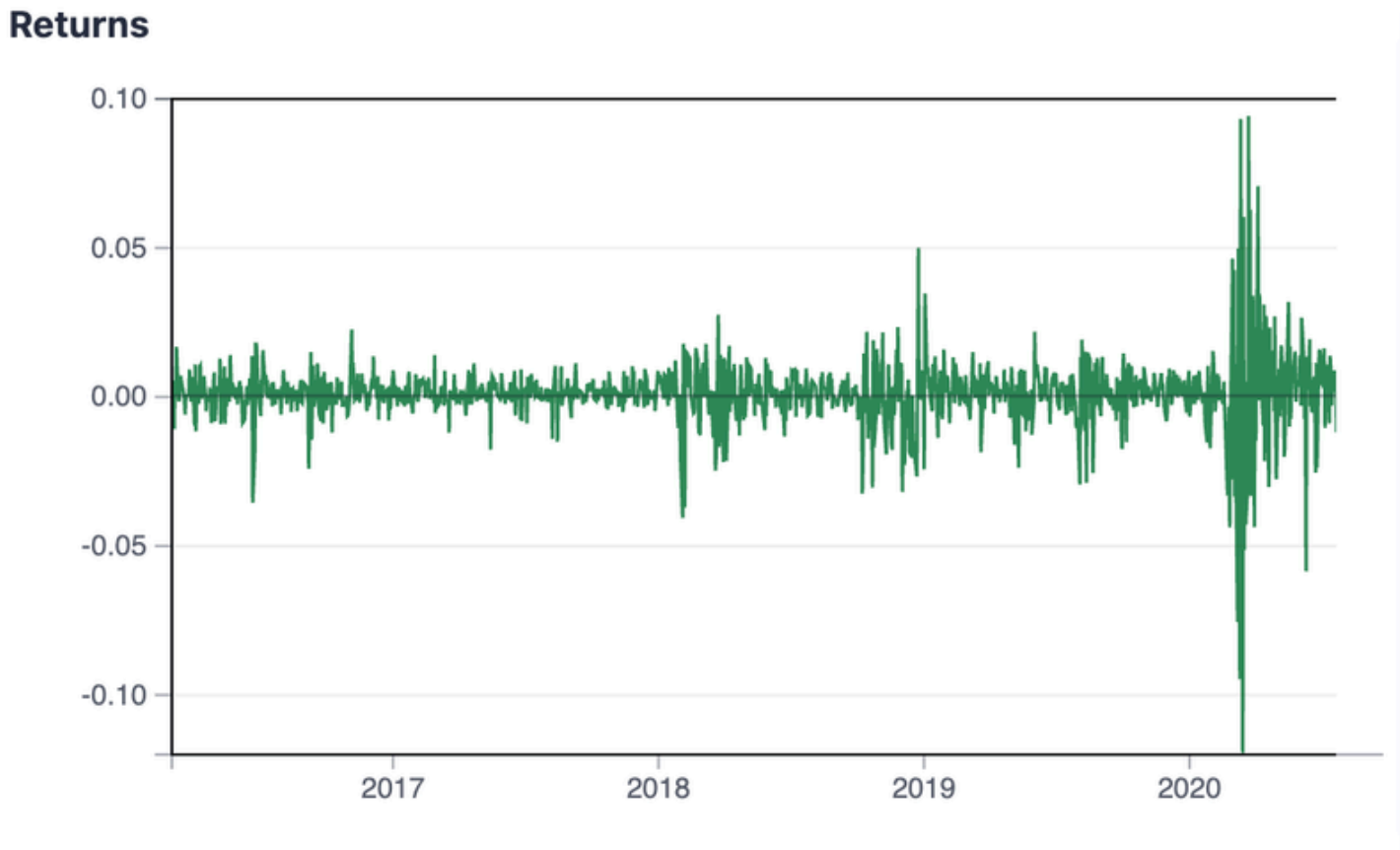




Price Chart Graph

For our first graph, we chose to create the standard price graph, which is the date on the x-axis, and the closing price of the corresponding day on the y-axis. Since the dataset contains 10 years of historical data, we chose to display all of it to provide as much context as possible about the historical trend of the S&P 500. The closing price is the blue line. There is also a yellow line, which is the average of the last 50 days of closing prices, and a purple line, which is the average of the last 200 days of closing prices. A golden cross (50 above 200) can symbolize a bullish trend indicator, while the opposite can be considered a bearish indicator.

Returns Graph

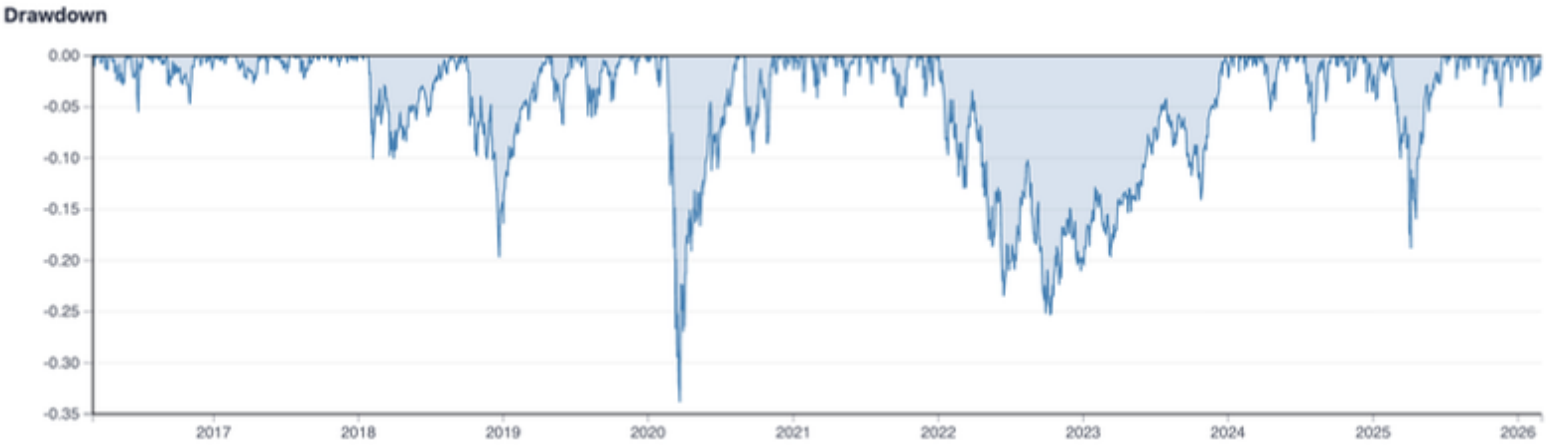


The returns graph shows the daily change of the S&P 500. This shows how much the index moved each day as a percentage. If the line is above 0 for the corresponding day, that means that the index went up for that day. If the line is below 0, that means the index went down for that day. This graph is the most useful for showing the short-term volatility of the index.

Volatility Graph

The volatility graph shows how unstable the market is. Volatility measures how big price swings are. This graph basically takes the standard deviation on the previous 30 days of returns, and analyzes how far apart each datapoint is. The standard way volatility is reported in finance is annualized volatility, which involves multiplying by the square root of 252 (which is typically approximately how many days the market is open per year). A volatility of 5-10% is considered a calm market, where a value of 25-40% is considered very volatile.

Drawdown Graph



The drawdown graph measures how far the index has fallen from its previous highest point (peak). For example, if the graph is at 0%, that means the index is at a new high. If the line is below that, it shows how far the stock has dipped. The graph is also shaded, indicating the size of the losses. Drawdown graphs are useful, as losses are very pronounced, allowing investors to identify risk and potential crashes. A drawdown of -5% might be considered a small dip, where something like -20% is considered a bear market.

What our Visuals Told us about the Data

Our visualizations helped us understand the basics of the stock market in a way that was easy for beginners in our group to grasp. One of the key takeaways was how significantly time impacts market behavior. An examination of the price chart shows long-term upward and downward trends, but also the smaller fluctuations that happen over shorter time frames. The moving averages were effective for illustrating longer-term trends, clearly showing the extent to which the market recovered after downturns.

The returns graph provides insights into market movements daily. While most daily movements are on the small side, there are large spikes representing surprises in either direction from time to time. These spikes tend to coincide with times of heightened volatility. The graph of volatility highlights specific periods of heightened volatility in the financial markets, which often coincide with major economic or financial events. In such times, the price action becomes unpredictable, making them riskier investments, suggest experts.

Finally, the drawdown visualization conveyed how far the market had declined from its previous peak at any point in time. This enabled us to gain perspective on just how far the market can decline in value and how long it can then take to recover to its peak levels. In sum, these charts helped us to visualize the market over time and understand how trends, risk, and recovery appear in actual financial data.

Answering our Questions

By creating several visualizations that each highlight a significant facet of stock market behavior, we were able to provide answers to our questions. We were able to comprehend how the market shifts over extended periods of time and how trends evolve over time, thanks to the price chart. In order to better understand short-term market movements, we also looked at daily returns, and the volatility visualization helped identify times when the market was especially erratic. Lastly, the drawdown graph helped us comprehend the size of market downturns by showing us how far the market had dropped from earlier peaks.

Since it displays multiple significant market indicators on a single dashboard, our visualization functions well overall. Additionally, the dashboard is interactive, which increases its analytical effectiveness. For instance, the price chart's selected time range is automatically applied to the other charts. This makes it simple for users to concentrate on a particular time frame, like 2018–2019, and observe how price, returns, volatility, and drawdowns all behaved during that time.

Future Remarks

Our visualization could be enhanced in a number of ways in the future. Adding annotations for significant financial or economic events is one potential improvement that could help explain abrupt market increases or decreases. Adding more filtering options, like the ability to choose particular years or time periods without having to drag the chart by hand, would also be an improvement. The dashboard would be even more helpful for examining and evaluating market behavior with these additions, which would give the data more context.